

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1.A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer:

1. Introduction

Artificial Neural Networks (ANNs) are widely used for predicting student performance because they can learn complex relationships between input features (attendance, study hours, previous grades, assignments, etc.) and output scores. The performance of a neural network is commonly evaluated using Mean Squared Error (MSE).

In this problem:

- Training MSE = 0.85
- Testing MSE = 5.20

The large difference between these two values indicates that the model performs well on training data but poorly on unseen test data. This situation generally indicates overfitting, where the neural network memorizes the training data instead of learning generalized patterns.

2. Mean Squared Error (MSE)

Mean Squared Error measures the average squared difference between the actual and predicted values.

Formula

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where,

- y_i = Actual value
- $\hat{y}_i$ = Predicted value
- n = Number of observations

A lower MSE indicates better prediction accuracy.

3. Error Identification

The model has

- Training MSE = **0.85**
- Testing MSE = **5.20**

The testing error is much larger than the training error.

This clearly indicates

Overfitting

The neural network has learned the training examples very well but fails to generalize to new unseen data.

The model remembers the training dataset instead of learning the underlying relationship between the input features and student scores.

4. Error Analysis and Reasoning

Several reasons may cause such a large difference between training and testing errors.

a) Model is Too Complex

If the neural network contains too many hidden layers or neurons, it becomes highly flexible. Instead of learning useful patterns, it memorizes the training dataset.

As a result,

- Training error becomes very low.
- Testing error becomes very high.

b) Insufficient Training Data

If only a small number of student records are available, the neural network cannot learn the actual distribution of data.

Instead, it memorizes the available samples.

Therefore, prediction accuracy decreases for new students.

c) Noise in Training Data

The dataset may contain

- Incorrect student marks
- Missing values
- Outliers
- Measurement errors

The neural network learns these noisy patterns, reducing its ability to generalize.

d) Too Many Training Epochs

Training for excessive epochs causes continuous reduction in training loss.

However,

after a certain point,

the validation and testing errors start increasing.

This is another indication of overfitting.

e) Lack of Regularization

Without regularization, the network weights become excessively large.
This increases model complexity and causes poor generalization.

f) Irrelevant Features

Features unrelated to student performance confuse the model.
Examples include

- Student ID
- Registration number
- Random attributes

These increase complexity without improving prediction.

5. Interpretation of Results

The large gap

0.85 → 5.20

shows that

the neural network is unable to generalize.

It performs excellently on training data

but

fails to predict unseen student scores accurately.

Therefore,

the model cannot be considered reliable for real-world student score prediction.

6. Corrective Measures

Several methods can reduce overfitting and improve testing performance.

1. Regularization (L1 & L2)

Regularization penalizes large weights.

Benefits:

- Reduces model complexity
- Prevents overfitting
- Improves testing accuracy

2. Dropout

Dropout randomly disables neurons during training.

Advantages

- Prevents neuron dependency
- Improves generalization
- Reduces overfitting

3. Early Stopping

Monitor validation error.

Stop training when validation error starts increasing.

Benefits

- Prevents memorization

- Saves computation time

4. Increase Training Data

Collect more student records.

Larger datasets help the network learn actual relationships instead of memorizing examples.

5. Feature Selection

Remove irrelevant attributes.

Keep only useful features such as

- Attendance
- Internal marks
- Assignment scores
- Previous semester GPA

This reduces complexity.

6. Cross Validation

Use k-fold cross-validation.

It evaluates the model using multiple data splits.

This produces a more reliable estimate of performance.

7. Data Normalization

Normalize input features.

Benefits

- Faster convergence
- Stable training
- Better prediction accuracy

7. Comparison Between Training and Testing Errors

Training Error	Testing Error
Very Low (0.85)	High (5.20)
Learns training data	Predicts unseen data
High accuracy	Low accuracy
Indicates good fit	Indicates poor generalization
Overfitting if gap is large	Needs improvement

8. Real-World Example

Suppose a teacher trains an AI model using marks of 100 students.

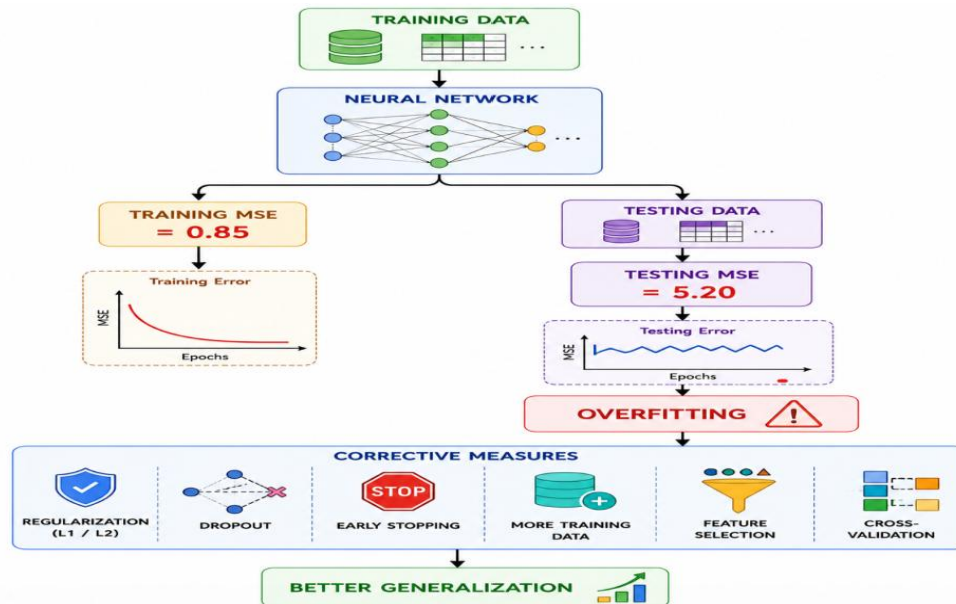
The neural network memorizes the marks of those students.

When marks of new students are given,
the predictions become inaccurate.

Hence,

Training MSE remains low,
while Testing MSE becomes high.

This is a practical example of overfitting.



10. Conclusion

The neural network exhibits overfitting, as indicated by the significant difference between the training MSE (0.85) and testing MSE (5.20). The model has learned the training data well but cannot generalize to unseen student records. Applying techniques such as regularization, dropout, early stopping, feature selection, cross-validation, data normalization, and increasing the training dataset can reduce overfitting and improve the model's prediction accuracy. These corrective measures help the neural network produce reliable and accurate results for real-world student score prediction.

Code of Conduct:

I Guggilla Venkatsai(192425388) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided